

Matias Freire

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EDUCATION

Stevens Institute of Technology

Hoboken, NJ

Bachelor of Science in Computer Science, Minor in Math

Sept 2024 – May 2028

Relevant Coursework: Algorithms, Data Structures, Discrete Structures, Computer Architecture and Organization, Multivariable Calculus, Vectors and Matrices, Probability, Mechanics, Biology and Biotechnology

Awards: STEP Alumni Scholarship: Selected as 1 of 3 recipients out of 200+ members for outstanding leadership

Certifications: Hugging Face AI Agents Fundamentals, Bloomberg Market Concepts

PROJECTS

U-Net MoS Image Synthesis & Analysis Platform | *PyTorch, Dash-Plotly*

- Developed an intuitive end-to-end Dash/Plotly platform for deep learning research of STEM defects in MoS₂, featuring a streamlined workflow that automates synthetic image generation with customizable defects, applies realistic noise injection, and implements a ResUnet model for vacancy and polymorph classification
- Studying nanomaterial research papers covering MoS, defect formation and characterization, and deep learning preprocessing and analysis techniques for biweekly/monthly presentations.

Interactive-Neural-Network-Digit-Classifer | *PyTorch, Streamlit, Numpy, Pygame*

- Achieved 96% accuracy on MNIST digit recognition by building and training a 4-layer Convolutional neural network (512-256-128-64 neurons) in PyTorch.
- Enabled real-time user interaction and visualization by integrating a PyGame drawing canvas and deploying the model with a Streamlit web interface

BAMM Canvas - HackRU 2025 | *React, Tailwind, Typescript*

- Implemented the front-end interface with React and TypeScript, ensuring a responsive and visually engaging UI.
- Designed and styled UI components using Tailwind CSS for a consistent and modern look, created an interactive Canvas for user input, and integrated real-time updates with backend predictions.

EXPERIENCE

Deep Learning Research Assistant/Intern

Aug 2025 – Present(ongoing)

Advanced Quantum Materials Lab

Hoboken, NJ

- Researched MoS defect characterization for deep learning studies and applied established preprocessing, synthetic data generation, and modeling techniques to analyze vacancy and polymorph defects within a unified research Dash-Plotly Hub.
- Synthesized and presented nanomaterials research to Dr. Eui-Hyeok Yang and others in the Lab through bi-weekly presentations, enhancing team-wide understanding of material properties and defect analysis methodologies.

Machine Learning Ignite Fellow Financial Track

Aug. 2025 – Present(ongoing)

AI4ALL

Remote

- Developing a financial forecasting model by applying topological data analysis to extract persistent features from time series data, capturing multi-scale non-linear market patterns, improving prediction accuracy over conventional statistical methods
- Studied supervised and unsupervised learning techniques under industry mentorship and presented results to 30+ peers and mentors at a student symposium.

LEADERSHIP ROLES & PROFESSIONAL DEVELOPMENT

Head of Extern Relations & Co-Chair of Technology

Apr 2025 – Present(ongoing)

Colorstack Stevens Chapter

Hoboken, NJ

- increased active membership by 50% and online engagement by 25% from Fall '24 to Fall '25 by spearheading social media strategy and organizing 14+ events per semester.
- Strengthened member coding proficiency by collaborating with peers to solve 32+ data structure problems each term

STEP Tutor-Counselor/Resident Assistant (TCRA)

Jun 2025 – Aug 2025

STEP Bridge Summer Program

Hoboken, NJ

- Strengthened STEM readiness for 50+ incoming freshmen by leading evening Calculus I/II and Physics recitations with targeted problem-solving support to build technical proficiency and conceptual understanding in fast-paced, credit-bearing courses.
- Provided personalized Calculus tutoring to 25 plus students weekly, including late night academic support, while supporting residential life operations as a Resident Assistant

TECHNICAL SKILLS

Programming Languages: Java, Python, C/C++, JavaScript

Web Development: TypeScript, React, Tailwind, Next.js

Developer Tools: Git, GitHub, VS Code, Jupyter Notebooks

Libraries: pandas, NumPy, Matplotlib, PyTorch, scikit-learn, Streamlit